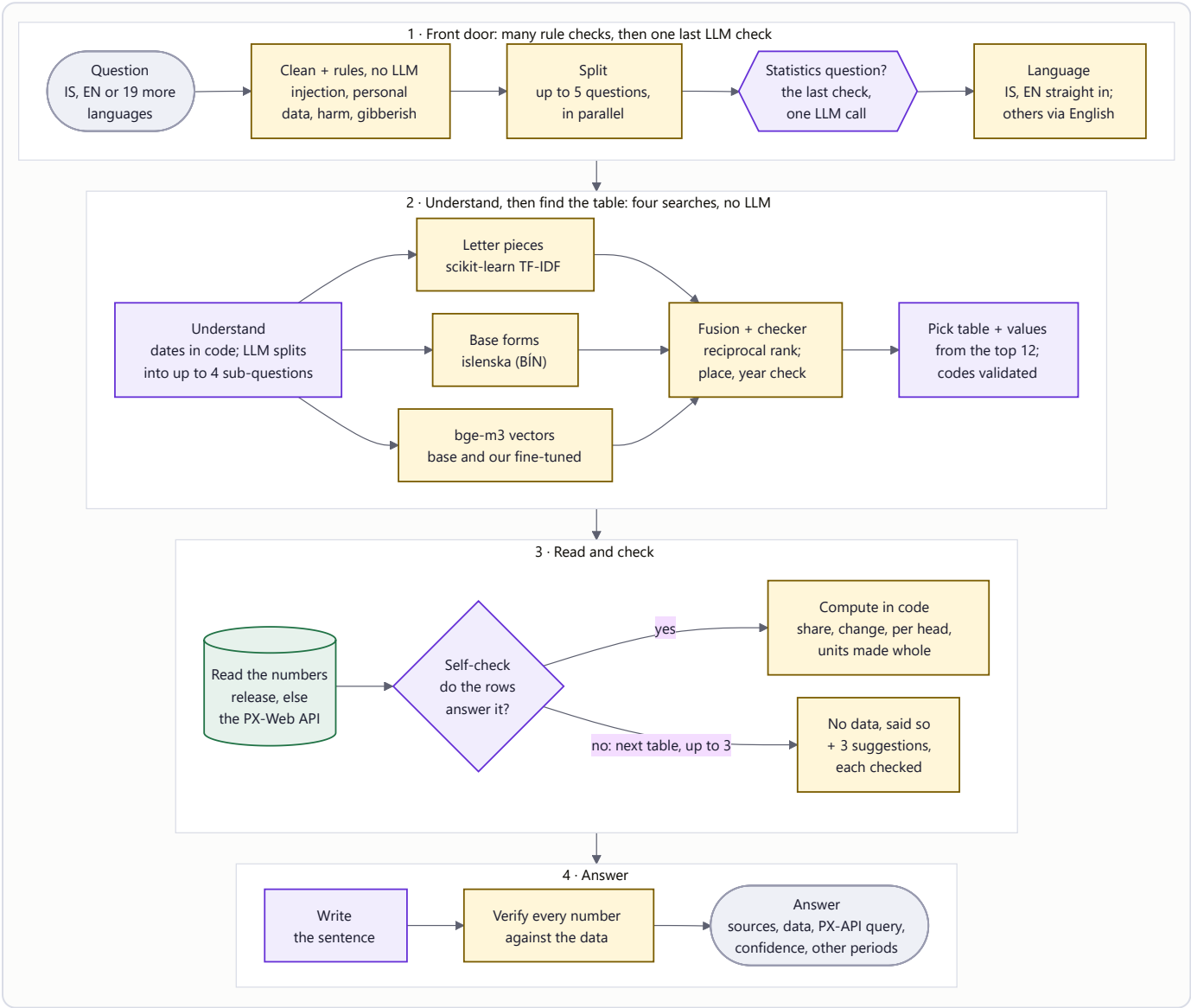
Spurðu Hagstofuna (Ask Statistics Iceland)

Project description and results · Gagnaþon 2026, Track 2 (direct PX-API) · Team: Atli and Soukaïna, who built it together · Code: github.com/Atlimh1/Tjarna

What we set out to achieve

A chatbot anyone can ask about Statistics Iceland's figures, in Icelandic, in English or in another language, that answers with the official number, its unit and period, and the table it came from, and that never invents a number. One principle runs through the design: **the model chooses, code does the rest**. The language model picks the table and the values and writes the sentence; code finds the candidate tables, reads the numbers, does every calculation, checks every number in the answer against the data and cites the source, with the exact PX-API query that reproduces it.

How it works



Architecture of one question. ☐ language model (Gemma 4 31B) ☐ code ☐ data ☐ question and answer
Runs on: vLLM serving Gemma 4 31B on 4 A100 GPUs on the JURECA supercomputer (Jülich), where our search model was also fine-tuned; the pipeline, a FastAPI service and the web page on a Raspberry Pi 5 or a JURECA login node.

- **Front door: many rule checks, then one last LLM check.** A short follow-up is rewritten with the question before, and the text is cleaned (broken encoding, pasted text; 500 characters at most). Rules with no model stop injection, personal data, harm, abuse, gibberish and plain sums; a message about self-harm gets the helpline numbers. A message is split into up to 5 questions, answered in parallel. For each, after the rules, one LLM call is the last check: is it a statistics question at all? Our table picker always picks a table (that answers more real questions right), so without this an off-topic message would get a number from some table; the rules stopped 0 of 16 off-topic inputs, and a cut-

off on the search score could not tell them apart. Then Icelandic and English go straight in; other languages go through English, with every number checked twice.

- **The data (Track 2).** The organizers' release, 1,886 tables in Icelandic and English as of 18 Oct 2026, and Statistics Iceland's public PX-Web API. No question comes with its table. A crawler walked the API's folder tree, one request a second, and saved every table's path, titles and value codes. Numbers are read from the release, else from the live API, and a button re-reads them live to say whether a figure is still the newest.
- **Finding the table, with no language model** (the starter kit's weak spot). Four searches over one index of both languages: scikit-learn TF-IDF on 3 to 5 letter pieces (typos, word forms); Icelandic base forms from BÍN (islenska) with synonyms and place names; bge-m3 vectors (sentence-transformers), computed once per table and compared by cosine similarity; and a bge-m3 we fine-tuned on JURECA with 4,580 questions that Qwen3-30B wrote for 924 tables (21 minutes on 4 A100s). Reciprocal rank fusion joins them, English questions are also searched in Icelandic, and a checker moves down tables that lack the place, country or year asked. The top 12 go to the model.
- **The model chooses, code checks.** The model picks the table and the value codes; code drops codes the table does not have and fills what the question left open with the newest period or the total, says so in the answer ("Miðað við: Ár 2026") and offers other periods or places as buttons. Dates like "í fyrra" become years in code. A self-check asks whether the rows answer the question; if not, the next table, up to three, else "no data". A year not published yet gets the newest figure, said so; a look-alike table that holds another figure is named under the answer.
- **Several tables.** The model splits a question into up to four sub-questions, one table each, read in parallel; anything beyond four is named in the answer, never dropped.
- **Code computes and checks.** Differences, percent changes, shares, ratios, rates per 1,000, sums, averages and whitelisted multi-step formulas, with units made whole first (millions of krónur per inhabitant). After writing, code checks every number in the answer against the data; if one fails, the rows are shown as they are.
- **No data, said plainly.** A subject map (the release's 155 folders, each with everyday words from one LLM call, 148 words corrected by hand) adds the subject's tables; the model writes up to six close questions, each is asked for real, and the first three that answer are offered as buttons.
- **Every answer shows** its tables and values, the unit, the table's own note, a citation, the PX-API query, a confidence measured on our tests and the response time.

Results on Statistics Iceland's data

Benchmarked with the organizers' own `gagnvist evaluate`, unchanged: an answer is right only if it states every expected number and cites a table that holds it. Expected numbers are read from the data release by code, not typed.

Question set	What it is	Right
Organizers' 60	their test questions: easy, medium (calculation), hard (two tables); Icelandic and English	**60 / 60**
Set 1 (frozen)	41 new questions we wrote before any fix aimed at them	**39 / 41**
Set 2 (frozen)	45 questions written by a separate AI agent from the catalog; never tuned on	**37 / 45**
real2 (frozen)	42 questions typed by us the way people type: typos, five languages, traps	**33 / 42**
dev3	62 questions over 61 tables no other set uses, written blind on 1 Nov (typos, relative dates, places)	**45 / 62**
Should be declined	questions Statistics Iceland does not publish (earthquakes, a firm's profit, forecasts)	**27 / 28**
Other languages	19 languages, 2 questions each: the same numbers as the English original	**37 / 38**
Guard	bad inputs stopped (injection, gibberish, harm, personal data, off-topic); real questions wrongly refused	**165 / 165; 2 of 1,008**

What each step adds (the organizers' 60, Gemma 4 31B; each rung adds one part of the system):

Model alone	Embedding search, like the starter kit	Our search	+ picking the table	+ calculation in code	+ sub-questions	Full system
0	**7**	**16**	**22**	**38**	**58**	**60**

Baselines and speed. The starter kit answered 7 of its own 52 test questions, and the organizers' pipeline run with our Qwen3-30B 8 of the 60. Ours takes about 4 to 5 s per answer; 10 questions sent at once were all answered in 19.4 s, with no errors. The 60 shaped many of our rules, so the frozen and blind sets (73% to 95%) are the fairer picture of new questions.

Limits

- Look-alike tables that count differently (dental care paid by households or by all private payers): the answer names the other figure, but its first choice can be the less fitting table.
- Some questions have two official figures (turnout by two definitions); years before 1998 split by place are rarely found (Reykjavík in 1950).
- The release is the snapshot of 18 Oct 2026, with a live check; the model runs as a booked job on a supercomputer, and the Raspberry Pi runs three of the four searches (no GPU there for our fine-tuned model).

What we would do next

- Test on real questions from real people, and fine-tune the search with them and with more English questions.
- Tell look-alike tables apart better: table cards with each table's notes made the picker worse when we measured them (the 60 fell from 60 to 54), so the notes are shown to the user under each answer instead.
- Run on smaller hardware (a single GPU or a DGX Spark) with a smaller model.

AI tools used

- **Claude Code** (Anthropic, Claude models) as a programming assistant under our direction: writing and refactoring code, running and analysing experiments on JURECA, building evaluation sets (set 2 and dev3 were written by separate agents that never saw the pipeline), and drafting documentation. Every change was measured on held-out question sets before it was kept, and we reviewed and decided on each design choice.
- **Claude** (claude.ai chat) to plan the work and get second opinions on our choices.
- **Models inside the product**, all open-weight and free to use: Gemma 4 31B (answers), Qwen3-30B-A3B (second opinion, and the questions our search model was fine-tuned on), bge-m3 and our fine-tuned bge-m3 (search), with GreynirCorrect and BÍN for Icelandic spelling and base forms.

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